QP CODE 25901388 **SEMESTER 4**

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2026
MG4DSCMAT204 – MATHEMATICAL TECHNIQUES IN BUSINESS ANALYTICS

Part A

(Any 5 questions. Maximum 2 marks can be earned from each question)

1. Explain the first order finite differences. Correct definition / formula with notation $\Delta y = f(x+h) - f(x)$ 2 Marks

(Give 1 mark for a suitable example or partial explanation) 1 Mark

2. Define Interpolation. Correct definition 2 Marks

3. What is meant by Inverse Interpolation? Correct definition / explanation 2 Marks

4. How forward interpolation differs from backward interpolation? Correct comparison / key differences stated 2 Marks

5. Solve $x^2 + 4x - 12 = 0$. Correct factorisation or quadratic formula applied 1 Mark

Getting correct answers $x = 2$ and $x = -6$ 1 Mark

6. Sketch the graph of $P = -x^2 + 8x - 12$; find vertex and intercepts. Correct vertex found: $x = 4,\ P = 4$ 1 Mark

Correct x-intercepts ($x = 2$, $x = 6$) and y-intercept ($P = -12$) identified 1 Mark

7. Find consumers' surplus if $P = 120 - 4Q$, $Q_0 = 6$, $P_0 = 96$. Setting up the integral $\int_0^6 (120 - 4Q)\, dQ - 96 \times 6$ 1 Mark

Correct answer $CS = 72$ 1 Mark

8. What is meant by Constrained Optimization? Correct definition 2 Marks

Part B

(Any 5 questions. Maximum 6 marks can be earned from each question)

9. Following the pattern in the table, predict the value of y for $x = 40$. Constructing the difference table correctly 2 Marks

Applying Newton's forward formula correctly 2 Marks

Getting the correct predicted value $y = 105$ (or appropriate value) 2 Marks

10. Find the value of $f(1.5)$ using Newton's formula from given data. Setting up

the forward difference table **1 Mark**

Writing Newton's forward interpolation formula correctly **1 Mark**

Correct substitution ($h = 1$, $p = 0.5$) **2 Marks**

Getting correct answer $f(1.5) = 44.5$ **(or approximate value)** **2 Marks**

11. Calculate $f(3)$ **by applying Lagrange's formula to the given points. Writing Lagrange's interpolation formula correctly** **1 Mark**

Correct substitution for each term **3 Marks**

Getting correct answer $f(3) = 9$ **2 Marks**

12. Price-demand relationship $p = \dfrac{200}{x+5}$**: draw graph and identify asymptotes. (a) Drawing the graph of** p **versus** x **correctly** **2 Marks**

(b) Identifying vertical asymptote $x = -5$ **and horizontal asymptote** $p = 0$ **2 Marks**

Economic interpretation of the asymptotes **2 Marks**

13. Business earnings: monthly salary 12,000 + 25 per item; total 17,000. (a) Forming correct linear equation: $12000 + 25x = 17000$ **2 Marks**

(b) Solving the equation **2 Marks**

Getting correct answer $x = 200$ **items** **2 Marks**

14. A company's total cost passes through $(10,\ 5000)$ **and** $(30,\ 11000)$**. (a) Finding slope** $m = \dfrac{11000 - 5000}{30 - 10} = 300$ **2 Marks**

Writing equation of cost line: $C = 300x + 2000$ **2 Marks**

(b) Slope = 300/unit (variable cost); y**-intercept = 2000 (fixed cost); correct economic interpretation** **2 Marks**

15. $C(x) = 120 + 5x$ **and** $C(y) = 200 + 8y$**; form total cost function. (a) Forming** $C(x, y) = 320 + 5x + 8y$ **correctly** **2 Marks**

(b) Substituting $x = 6$ **and** $y = 4$ **2 Marks**

Getting total cost $= 382$ **2 Marks**

16. If marginal cost $MC = 2x^3 + 5x + 12$**, find additional total cost when output increases from 2 to 4. Writing** $\int_2^4 MC\,dx = \int_2^4 (2x^3 + 5x + 12)\,dx$ **2 Marks**

Integrating correctly: $\left[\dfrac{x^4}{2} + \dfrac{5x^2}{2} + 12x\right]_2^4$ **2 Marks**

Getting correct answer $= 94$ **(or approximate value)** **2 Marks**

Part C

(Any 3 questions. Maximum 10 marks can be earned from each question)

17. Cubic polynomial $y = x^3$: how many columns of differences are needed? Verify with given table. Stating that 3 columns of differences are needed for a cubic polynomial **2 Marks**

First differences (Δy) calculated correctly **2 Marks**

Second differences ($\Delta^2 y$) calculated correctly **2 Marks**

Third differences ($\Delta^3 y$) showing constant value ($= 6$) **2 Marks**

Correct conclusion: constant reached at 3rd column; proper verification **2 Marks**

18. Evaluate $f(3.2)$ by constructing a backward difference table. Constructing the backward difference table correctly **3 Marks**

Writing Newton's backward interpolation formula **2 Marks**

Correct substitution ($x_n = 4,\ h = 1,\ s = -0.8$) **2 Marks**

Getting correct answer $f(3.2) \approx 13.4$ (or appropriate value) **3 Marks**

19. Find quadratic polynomial $P(x)$ and solve $P(x) = 0$ using polynomial interpolation. Setting up the difference table for given data **2 Marks**

Applying Newton's forward interpolation formula **3 Marks**

Writing out the quadratic polynomial $P(x)$ correctly **2 Marks**

Solving $P(x) = 0$ and getting correct roots **3 Marks**

20. Define central difference operator δ; construct table; find missing value at $x = 0.5$. Correct definition of central difference operator δ **2 Marks**

Constructing the central difference table for given data **3 Marks**

Identifying and applying the appropriate formula for $x = 0.5$ **3 Marks**

Getting the correct missing central value **2 Marks**

21. Profit $P = -2x^2 + 40x - 120$: draw graph, find vertex and intercepts. (a) Drawing the graph of P versus x correctly **3 Marks**

(b) Finding the vertex: $x = 10,\ P = 80$ **3 Marks**

(c) Finding x-intercepts ($x = 3$ and $x = 17$) and y-intercept ($P = -120$) **4 Marks**

22. Producers' surplus: $P = 4Q^3 + Q^2 + 2Q + 6$ at $Q_0 = 2$, $P_0 = 50$. Writing Producer's Surplus formula: $PS = P_0 Q_0 - \int_0^{Q_0} P(Q)\, dQ$ **2 Marks**

Correct integration of $\int_0^2 (4Q^3 + Q^2 + 2Q + 6)\, dQ$ **4 Marks**

Evaluating the definite integral: $\left[Q^4 + \dfrac{Q^3}{3} + Q^2 + 6Q \right]_0^2$ **2 Marks**

Getting correct answer $PS = 100 - 32.67 \approx 67.33$ **(or approximate)** **2 Marks**

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